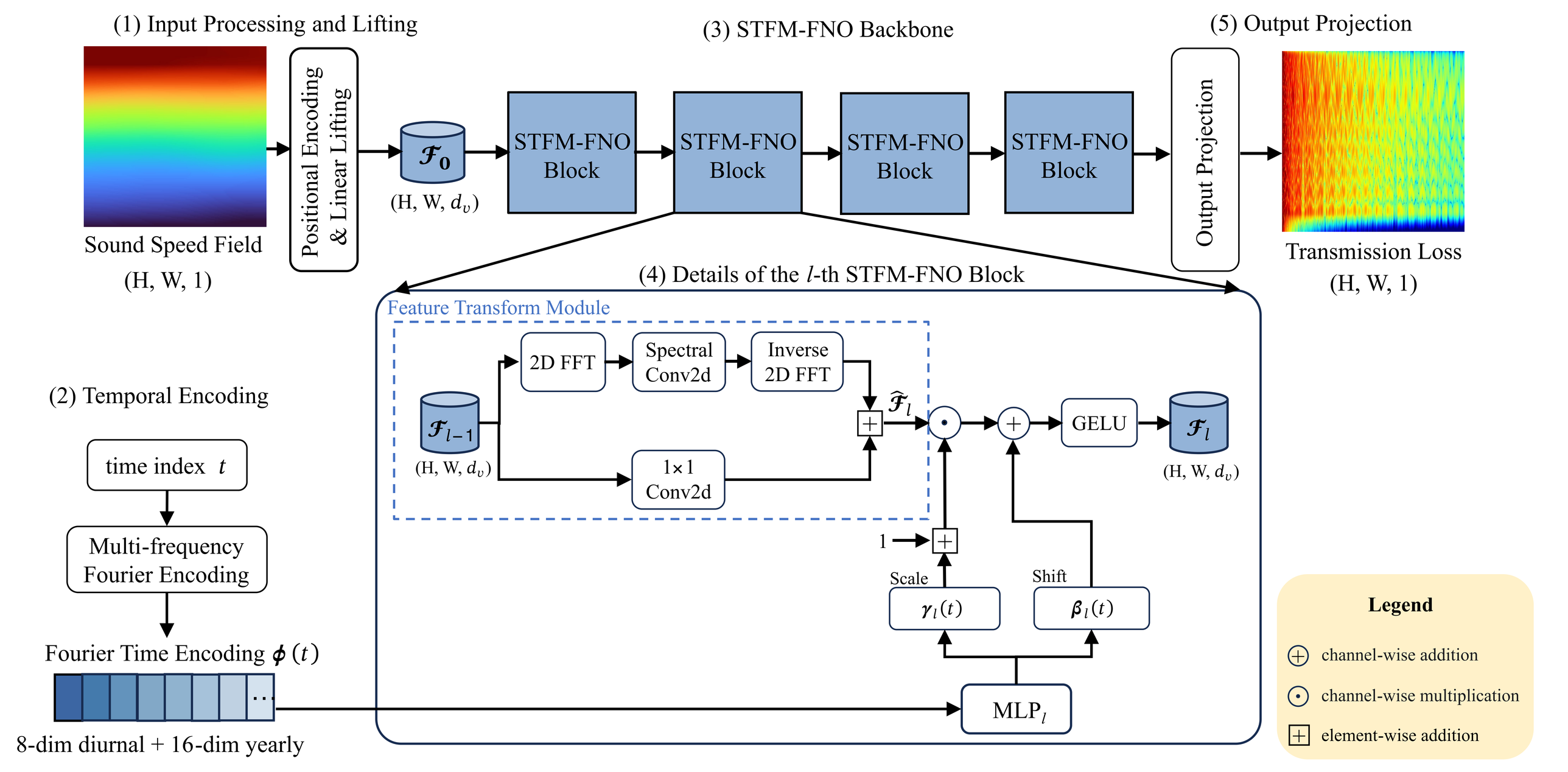

(1) Input Processing and Lifting
Sound Speed Field
(H, W, 1)
Positional Encoding & Linear Lifting
$\mathcal{F}_0$
(H, W, d_v)
(3) STFM-FNO Backbone
STFM-FNO Block
STFM-FNO Block
STFM-FNO Block
STFM-FNO Block
(5) Output Projection
Output Projection
Transmission Loss
(H, W, 1)
(4) Details of the l-th STFM-FNO Block
Feature Transform Module
2D FFT
Spectral Conv2d
Inverse 2D FFT
$\mathcal{F}_{l-1}$
(H, W, d_v)
1×1 Conv2d
$\widehat{\mathcal{F}}_l$
GELU
$\mathcal{F}_l$
(H, W, d_v)
1
Scale
$\gamma_l(t)$
Shift
$\beta_l(t)$
MLP_l
(2) Temporal Encoding
time index t
Multi-frequency Fourier Encoding
Fourier Time Encoding $\boldsymbol{\phi}(t)$
8-dim diurnal + 16-dim yearly
Legend
channel-wise addition
channel-wise multiplication
element-wise addition